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Chapter 3 Natural Language Processing for Industrial and Systems Engineering

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3.1 Introduction

Recent advances in natural language processing (NLP) are transforming the world we live in and the ways in which we work. NLP technology has been productively used in many domains, including industrial and systems engineering (ISE), for decades, e.g. to find relevant documents in large corpora, translate texts, extract information from documents, or interact with systems in a more natural way. Technological advances, as well as the increased availability of data and computing power, have led to an almost ubiquitous availability of powerful large language models (LLMs) through AI-as-a-service offers like the GPT models of OpenAI or

open-source alternatives like BLOOM (Le Scao et al. 2023) or Llama (Touvron et al. 2023). These new models and systems replace not only traditional rule-based NLP and expert systems, but also less advanced stochastic approaches.

LLMs, and thereby NLP more broadly, have gained significant public attention for showing capabilities that were previously thought to be displayed only by humans in a wide range of domains, from medicine (Kung et al. 2023) to law (Katz et al. 2024). Engineering is no exception in this regard: Pursnani et al. (2023), for example, have shown that LLMs like ChatGPT are able to pass the US fundamentals of engineering exam, indicating potential capabilities to solve complex problems in the domain of engineering. While we should be critical about the practical significance of such isolated achievements in rather artificial settings (and often also the conditions under which they have been achieved), there is no doubt that the capabilities of NLP models are evolving rapidly and will most likely continue to do so in the foreseeable future. These new powerful models will enable a wide variety of applications of NLP in ISE that were not possible before and thereby support, and sometimes replace, human expertise and labor. Equipped with vast knowledge from data and good language understanding capabilities, NLP system will be able to streamline and support various processes within industrial systems, from supply chain management to production scheduling. They will help to extract insights from unstructured data sources like maintenance logs and customer feedback and thereby enable engineers to make faster and better data-driven decisions. As the technology continues to improve, NLP systems will become indispensable tools for driving innovation and transformation in ISE. Skills in configuring, using, supervising, and assessing such systems will be vital in the future for engineering professionals and researchers.

Despite the already achieved technical quality, the current generation of NLP systems still suffers from a number of shortcomings, e.g. with regard to the factual correctness of their outputs and their ability in logical reasoning and planning. In a high-risk application domain, like engineering, these shortcomings still limit the practical applicability in settings without a so-called human-in-the-loop, i.e. human oversight of the system.

Nevertheless, existing NLP techniques can be and are already applied for a number of tasks within ISE and increasingly also ISE education. The described technologies will not only change the domain itself but also the people working in it. The current generation of engineering students already uses NLP technology as a natural tool, just like a calculator. At the latest when they graduate and enter the workforce, these tools will also become a normal sight in the workplace. This chapter aims to give an overview of how the recent advances in NLP can be applied to ISE and how NLP will continue to shape engineering practice. We first describe recent advances and trends in the field of NLP in general (Section 3.2), before describing domain-specific challenges and applications in ISE (Sections 3.3 and 3.4). We will conclude the chapter with an outlook on imminent developments and their potential impact on the domain (Section 3.5).

3.2 Advances and Trends in NLP

NLP is a highly dynamic field at the intersection of artificial intelligence (AI) and (computational) linguistics. While the predominant approach for decades, particularly in practical applications, has been rule-based systems (Chiticariu et al. 2013), they disappeared almost completely within the last 10 years to make space for stochastic systems that no longer need explicit representation of expert knowledge and instead learn from large amounts of data.

For a long time, there was also a distinction within the field of NLP between natural language understanding (NLU), i.e. processing text as input to derive structured information, e.g. for tasks like text classification, information extraction, or named entity recognition, and natural language generation (NLG), i.e. producing text as output from different inputs like data, images, or text, e.g. for tasks like translation, summarizing, or data-to-text. However, LLMs and their abilities as few- and zero-shot learners based on so-called prompts (see also Section 3.2.1) have reduced the significance of this distinction by unifying both aspects in one model.

Giving a comprehensive overview of all developments in NLP in just the last 10 years would be a forlorn endeavor. Therefore, this section will fo-

cus on three major trends (unsupervised approaches, large language models, and the use of knowledge graphs) that are particularly important for NLP applications within the context of ISE. For a more detailed deep-dive into developments in the field of NLP more broadly, a wide variety of overview papers exist, e.g. by Khurana et al. (2023), Min et al. (2023), and Fanni et al. (2023).

3.2.1 Large Language Model

The most prominent development in NLP in recent years arguably has been the development of LLMs like ChatGPT. Technically, ChatGPT and other LLMs build on the transformer architecture introduced by Vaswani et al. (2017). Like most stochastic language models, LLMs are trained on huge amounts of data, mostly crawled from the internet, and analyze the co-occurrence of words in order to derive a mathematical representation of the meaning of words, usually in the form of a vector. Before the popularization of the transformer architecture, approaches like Word2Vec (Mikolov et al. 2013) or GloVe (Pennington et al. 2014) used to calculate one static vector representation, a so-called word embedding, for each word. Particularly, for ambiguous words that can have different meanings in different contexts, which is often the case for domain-specific vocabulary, such a static representation is often insufficient. Transformer models, like BERT (Devlin et al. 2019), went away from such static representations. Instead, they analyze the meaning of a word within the context, often the sentence, it appears in. So instead of assigning each word one static vector, they assign each word a vector in the context it appears in. Thereby, these models are able to capture the, sometimes subtle, changes in the meaning of a word depending on the context it is used in.

As mentioned before, this approach has already been popularized in 2019. One of the most important differences between these language models of the previous generation, like BERT or T5 (Raffel et al. 2020), and what we now call “Large Language Models” is, as the name suggests, their size. The size of language models is often measured based on the number of parameters they have. The parameter count of a model represents the number of weights in the neural network that are adjusted during the training of the model. The more parameters a model has, the more “infor-

mation," or data about the relation between words, they can potentially store. Next to the amount of data that is used to train a model, the parameter count seems to be the most important predictor for the capabilities of a modern language model.

At the same time, the amount of parameters also heavily influences the computational power that is needed to train the model. The base model of BERT has 110 million parameters, the BERT large model has a total of 340 million parameters (Devlin et al. 2019). In comparison, the last open-source GPT-model, GPT-2, already has 1.5 billion (i.e. 1500 million) parameters (Brown et al. 2020). Llama 3, an open-source LLM of the, at the time of writing, latest generation, has up to 70 billion parameters, while GPT-4 is rumored to have more than a trillion parameters (Dwivedi et al. 2024).

Training such LLMs is an energy- and time-consuming task. Therefore, it is neither economically nor ecologically feasible to regularly re-train LLMs with new data. Therefore, most LLMs are based on what can be seen as outdated data. The models provided by OpenAI, for example, are mostly trained with data that has been collected before September 2021. If they are asked about recent events, for example, they therefore tend to give incorrect or incomplete answers. One approach to tackle the problem of outdated information is to allow a system to access the internet and incorporate it as external information (see also Section 3.5), like Microsoft's Copilot can do. However, this approach also opens up an attack vector for so-called prompt injection attacks. By hiding instruction to an LLM in the text of a website, or another document that is ingested by an LLM, an attacker can trigger the LLM to perform behavior that is unwanted and potentially dangerous to its user. In this way, attackers can prompt the LLM to produce incorrect outputs or extract data that a user feeds into the LLM (Greshake et al. 2023).

While LLMs in general have improved the performance of NLP systems in almost all relevant tasks, they also have some inherent limitations that are difficult to overcome. One of the main challenges is factual correctness. Under the term "hallucination," a number of different types of errors are summarized that, in essence, refer to the production of factually

incorrect texts by LLMs and other NLG models (van Deemter 2024). While the generation of factually incorrect output in itself is a problem, particularly for LLMs, another problem is the unpredictability with which they produce such output. Sometimes a very slight change in an input prompt can result in a very different result that is factually wrong but still looks credible on the surface. This unpredictability makes it very difficult to use the output of LLMs in a setting where no additional checks are available.

Like all stochastic models, LLMs are also prone to incorporate and replicate biases that are learned from the training data (Bender et al. 2021) and many studies have shown such biases in commercially available LLMs (Fang et al. 2024; Motoki et al. 2024; Urchs et al. 2023).

Another concern, particularly for hosted "as-a-service models," like ChatGPT, is data safety. Input prompts that contain private information about individuals, for example, might violate data protection regulations. Some companies also banned their employees from accessing such models in the workplace out of fear that sensitive company information, like trade secrets, might be put into such systems where it is not always clear how the operators will use this data.

3.2.2 Unsupervised Approaches

One of the main limiting factors in training machine learning (ML) models, next to the availability of computing power, is the availability of data. Supervised ML approaches, in particular, rely on the existence of annotated data, i.e. data that has been manually labeled by experts with relevant information. While computing power has become cheaper and, through services like Amazon's AWS or Microsoft's Azure, more easily accessible, the costs of expert annotation remain high and are rising. Having a sufficient amount of labeled data is, therefore, a limiting factor for the development of NLP models for domains like ISE. Alternative approaches, so-called unsupervised approaches, which do not rely on labeled data but are trained on unlabeled data, for a long time were not able to produce outputs that are as useful as supervised approaches. However, this has changed. By being trained on, previously unimaginable amounts of data, which are mainly crawled from the internet, LLMs, and

other unsupervised models, are able, without relying on labeled data, to complete a range of tasks they have not been specifically trained on. Particularly, for domains like ISE, where annotations need to be done by experts with a highly sought-after expertise, labeling data is a very expensive task. Such domains, therefore, particularly benefit from the new capabilities of models that were trained in an unsupervised fashion.

In addition to completely unsupervised approaches, so-called few-shot learning is another approach to reduce the amount of necessary (labeled) training data. Instead of training a model on hundreds, often thousands, of examples, few-shot learning uses as little as 1–64 labeled examples (Touvron et al. 2023). Language models, in particular, have shown to be good few-shot learners when given labeled examples as part of the input (or prompt) of the model (Brown et al. 2020). A completely unsupervised approach to translation could for example be the prompt "Translate the following word to French: house." Such a prompt would be called a zero-shot approach because no example is provided. A few-shot approach, on the other hand, would be the prompt: "Translate the following words to French: house => maison, window => fenˆetre, door =>."

While it is widely accepted that LLMs have good zero- and few-shot learning abilities, we should be careful about when to attribute these labels. For commercial LLMs, we do not know for sure on which data the models have been trained and therefore cannot be certain whether they were really not trained on a specific task. A study by Balloccu et al. (2024) has shown that the training data that was used to train the models GPT-3.5 and GPT-4 contains more than 4.7 million samples from 263 scientific benchmark datasets. That does not only mean that testing the models on these benchmarks would not be a few- or zero-shot test, but that the models were actually trained on the very samples they are now tested on, rendering the evaluation completely pointless.

3.2.3 Knowledge Graphs

Historically, ontologies and knowledge graphs have played an important role in applications of AI in engineering. Ontologies for Oil and Gas (Gulla 2009), industrial engineering (Poveda-Villalón et al. 2022; Terkaj et al.

2022), software engineering (Calero et al. 2006), systems engineering (Van Ruijven 2013), and many other engineering disciplines, have been used to create software and particularly expert systems, i.e. traditional AI systems that rely on explicitly encoded expert knowledge. Such expert systems can, for example, be used for the planning of industrial facilities (Kumara et al. 1987), manufacturing planning (Leo Kumar 2019), or decision making for technology acquisition in small- and medium-scale manufacturing organizations (Bayraktar and Gözlü 1994).

For some time, it looked like the success of purely stochastic, unsupervised approaches in NLP (see Sections 3.2.2 and 3.2.1) would make the use of explicitly coded knowledge, e.g. in the form of knowledge graphs, obsolete. However, more recently, knowledge graphs are experiencing a revival as a supplement to unsupervised approaches, such as LLMs, in many NLU and generation tasks (Schneider et al. 2022). Pan et al. (2023) presented an extensive overview of how LLMs and knowledge graphs can be combined and the opportunities that arise from such a connection, e.g. for the extraction of information from texts or increased text generation quality.

In the engineering domain, different proposals have been made to combine the two technologies in practical applications. Meyer et al. (2024), for example, suggested that LLMs could be used in combination with knowledge graphs to automatically generate SPARQL queries from natural language to query the knowledge base or also to assist with the construction of the knowledge graph itself. LIU et al. (2024) used a combination of an LLM and a knowledge graph in aviation assembly to detect faults in the assembly process. Their results not only suggest that the combination of both approaches can increase the accuracy of fault diagnosis and troubleshooting schemes in comparison to just using an LLM, but also that the combination can reduce the necessity to fine-tune LLMs and thereby significantly reduce the necessary computational load. Li and Starly (2024) presented a use-case for combining LLMs and knowledge graphs to identify manufacturing services and making decisions about which manufacturing service provider to choose. In the future, such combinations of structured knowledge and LLMs could improve their applicability, partic-

ularly in very domain specific tasks that are not represented well in the training data of LLMs.

3.3 Domain-specific Challenges in ISE

While most of the approaches that are used in modern NLP systems can be applied independently of the domain, and often even the language, every application domain brings its own challenges. In this section, we will discuss some of the most important domain-specific challenges in ISE and how they influence the applicability of NLP techniques within the domain but also which technologies can be used to address these challenges.

3.3.1 Domain-specific Vocabulary

Like other highly specialized domains, ISE uses a lot of domain-specific vocabulary, i.e. words that are not commonly used in everyday language or words that also exist in everyday language but are defined differently within the context of the domain. Domain-specific vocabulary is challenging because its meaning can often only be learned from domain-specific documents that might not be available in sufficient quantity or is at least not represented sufficiently in the training data of a general-purpose model. Especially, approaches that assign each word in the vocabulary one specific mathematical representation, independent of the context, like the aforementioned Word2Vec, often struggle to capture the meaning of domain-specific vocabulary well when trained on general data. Therefore, it has proven to be a successful strategy to train models on domain-specific data to increase their ability to handle domain-specific vocabulary. In the engineering domain, that has been shown by Braun et al. (2021) and Schopf et al. (2022).

Newer models, like BERT and the GPT-models, with their nonstatic representations, tend to be better at catching these different, domain-specific meanings even when trained on general data. Nevertheless, at least for smaller transformer models like BERT, studies in different domains like material science (Trewartha et al. 2022) and architecture, engineering, and construction (Zheng et al. 2022) have shown that models that have

been trained on smaller but domain-specific corpora tend to outperform models that have been trained on larger general corpora in domain-specific tasks.

While researchers already called for domain-specific LLMs in engineering (Pal et al. 2024), it remains unclear how such a model could be trained. BLOOM Le Scao et al. (2023), for example, was trained on 1.6 TB of text or 350 billion unique tokens, the so-called ROOTS corpus (Laurençon et al. 2022). Among this corpus are already vast amounts of engineering-related documents, e.g. from Wikipedia or proceedings from the European Parliament that contain many relevant regulations and norms. Therefore, it is not clear whether there would even still be significant amounts of domain-specific data immediately accessible that has so far not been used to train LLMs. Because the training of LLMs is so expensive, it has yet to be shown whether the positive effect of having more data available for training outweighs the advantages of having more domain-specific data in LLMs or whether they would still benefit from smaller, more specific corpora.

3.3.2 Multimodality

In the domain of ISE, one of the primary challenges lies in the diversity and complexity of the data that are used. Unlike some other domains, such as social media analysis or law, ISE data often includes technical specifications, sensor data, and operational logs, which can vary significantly in structure and format. Many relevant documents in the ISE domain, like norms, scientific articles, and data sheets, often contain not only texts but also drawings and visualizations that can contain important information. Analyzing not just the written text but also these figures and linking both modalities is still a big challenge for NLP techniques, although significant advances have been made recently.

Traditionally, most generative AI models, like ChatGPT, but also image generation models like Dall-E have one fixed modality for input and output. In case of ChatGPT, the modality for both input and output is the same (text), while for Dall-E and other image generators, the input is text and the outputs are images. Although image generation models can, in

theory, produce text (as part of an image), they are infamously bad at it (Marcus et al. 2022).

Only recently, models that can handle multiple modalities, i.e. multimodal LLMs, started to perform better. In their survey, Yin et al. (2023) have identified more than 50 multimodal LLMs. They conclude that "the development of m[ulti-modal] LLMs is still in a rudimentary stage" (Yin et al. 2023, p. 13). They identified some limitations they found across the different models, among them the fact that multimodal LLMs struggle to process long contexts like longer videos sequences or longer text sequences that are mixed with images. Particularly the limitation with regard to long texts with images is a very relevant restriction in engineering; however, one that is already being addressed by ongoing research (see Section 3.5).

3.3.3 Complexity

In many domains, from philosophy to medicine, NLP algorithms have to deal with complex input texts. We often anthropomorphize LLMs and other AI models by talking about them like we would talk about humans. We say an LLM "thinks" or "lies" although they do nothing like that. They calculate probabilities and generate texts based on these probabilities. But because we regularly anthropomorphize these models, we sometimes tend to also assume that what is complex to us is also complex to LLMs; however, that is not always the case. A very well-documented shortcoming of ChatGPT, but also other LLMs, is logical reasoning and mathematical thinking (Frieder et al. 2024). Both capabilities are particularly important in the engineering domain. Valmeekam et al. (2022) showed that the lacking ability in logical reasoning results in LLMs being unable to reliably generate plans, particularly under the constraint to optimize costs. So more than linguistic complexity, LLMs tend to struggle with mathematical and logical complexity, both of which are high in many documents that can be found in the domain of engineering. While the latest generation of LLMs, like GPT-4, already significantly improved in this area, it remains a challenge.

3.3.4 Liability and Accountability

The current generation of NLP systems already works very well in many cases. However, even if a system in general performs very well on a task, one of the main challenges is that it is very hard to predict for which cases the system will fail. Since an LLM by itself has no measurement for insecurity of a response, while in some domains, like customer support, it might be manageable if a system is wrong once in a while and the advantages it brings would still outlast the disadvantages, in domains where the stakes are high, the weighting of facts might lead to a different result. A wrongly predicted inventory need might halt a whole production and cost a company hundreds of thousands per day. A poorly designed machine that is not up to safety standards, e.g. because they were interpreted wrongly by an NLP system, can even endanger lives. Therefore, in such high-stakes environments, for the foreseeable future, whatever a system produces as output has to be evaluated and approved by a human. In the end, this human operator might be liable for whatever the machine suggests. Validating the output of, e.g. an LLM is often not trivial because of their complexity and the inability to explain how they derive a certain output. While it is possible to ask an LLM to provide an explanation, the text that is generated is not (necessarily) an indication of how the model actually derived its result but just another text it produces. One strain of research that tries to tackle this challenge is so-called retrieval-augmented generation in which an LLM is augmented with specific, curated documents on which it can base its reasoning and which it can also provide as a source, to simplify the process of validating the results (see also Section 3.5).

3.4 Applications of NLP in ISE

Despite the described challenges of the domain and existing limitations of the technology, NLP has already proven to be helpful for many applications in ISE. In the application of industrial maintenance alone, Zhong et al. (2024) identified, for example, 27 studies that have applied NLP to a wide variety of tasks, including fault analysis, decision support, maintenance knowledge extraction, and downtime prediction. In ISE more broadly, there is a wide variety of tasks that can be supported with

NLP systems, from domain-independent standard tasks like automatic text summarization (Klymenko et al. 2020) to very domain-specific tasks like man-hour prediction in the Engineering-Procurement-Construction (Kim et al. 2020a). The variety of possible applications is so large, that it would be impossible to discuss all of them in this context. Therefore, we want to focus on four application areas that are traditionally very strongly connected with human knowledge-work and therefore have been considered very difficult to support with NLP for a long time. However, with the previously described advances, we now see NLP systems being developed for all of these areas. The four areas of application are technology scouting, analysis of requirements documents, analysis of regulatory documents, and engineering education.

3.4.1 Technology Scouting

AI is just one rapidly developing technology that is relevant to ISE. Other such technologies and trends include, for example, 3D printing and virtual reality. Both of which have significant impact on, for example, manufacturing and engineering more generally. An important task for innovative companies and researchers alike is to keep up with the latest technological developments and assess how they affect their projects or products. With scientific and trade publications, trade fairs, news portals, and blogs as potential sources, keeping up with these developments and filtering important developments from “noise” has become harder than ever. Using NLP technology can help to identify relevant technologies faster and reduce the necessary amount of manual labor.

Unsupervised text classification approaches, like Lbl2Vec (Schopf et al. 2021, 2023a, 2023b,), can help to quickly identify if a certain text covers a topic that is relevant to the user by combining the advantages of text classification and topic modeling. Text classification is a supervised approach in which a text is classified in one of a number of predefined categories. Topic modeling, on the other hand, is an unsupervised approach by which texts are clustered into similar groups based on text features. The advantage of text classification approaches is that the categories are predefined ensuring that the classification is of practical relevance and helpful in identifying relevant texts. Topic modeling approaches, on the other

hand, have the advantage that they do not rely on annotated data; however, it is also not guaranteed that the resulting clusters are meaningful for the task at hand. Lbl2Vec combines the advantages of both approaches by not relying on annotated data and still classifying texts into predefined categories. In order to achieve that, the approach relies on a series of relevant keywords that have to be provided for each topic. Based on these keywords and the documents, the algorithm creates a joined embedding space that contains document vectors and topic vectors that are calculated for each topic based on the keyword list.

Zhou et al. (2019) presented an approach to automatically detect texts that describe emerging technologies. The approach combines new technologies, like semi-supervised clustering techniques, with more traditional techniques, like n-gram frequency or tf-idf. The resulting system is able to detect emerging technologies and automatically differentiate them from existing technology, supporting the ability of companies and researchers alike to quickly identify texts that describe novel technologies.

Beyond the simple detection of relevant texts, NLP technologies can also be used to extract more structured and semantically rich information from documents, e.g. by extracting relations between companies to automatically build an ecosystem of suppliers (Braun et al. 2018), or extracting new technological knowledge directly from patents (Wei et al. 2024; Yue et al. 2022). Such information can be used to further filter texts for technologies that are not only new but also relevant for specific use cases or problems. By now there is a number of companies that offer such technology scouting as a service based on NLP systems.

3.4.2 Analysis of Requirements Documents

The analysis of requirements is a crucial step in the engineering process, aiming to extract and understand the needs and specifications from textual descriptions. Over the years, NLP techniques have been increasingly employed to automate and enhance individual steps of this process. Zhao et al. (2021) gathered an extensive list of different steps and tasks NLP can be applied to. It can, for example, be used to assess the linguistic quality of requirement documents. Requirement documents should use clear and

unambiguous language. NLP can help to detect vague phrases, like “as soon as possible,” or weak verbs like “may.” More generally, the detection of “smells” in requirement documents, i.e. potentially problematic passages, independent of the underlying problem can be done by NLP systems (Veizaga et al. 2024).

Text classification techniques can be used to automatically classify requirements into categories like functional and nonfunctional requirements (Zhao et al. 2021). Systems that analyze requirements texts have, for example, been successfully applied in the railway domain (Rosadini et al. 2017). Arora et al. (2024) presented a system that can design test scenarios based on requirements. These are just some examples of how NLP can help in the analysis of requirements documents.

However, NLP can also help in requirements engineering more broadly. As described by Arora et al. (2023), applications of NLP are not limited to the processing of requirement documents but can be found in all stages of the requirements engineering process, including during requirements analysis, requirements validation, requirements elicitation, and requirements specification.

3.4.3 Analysis of Regulatory and Legal Documents

Keeping up with the latest regulations and ensuring compliance with technical regulations is a time-consuming task, especially in industries where products are designed for multiple markets with varying regulatory frameworks, like the automotive industry. NLP technology can help to analyze regulatory texts and ensure compliance. Xu and Cai (2021), for example, developed an ontology-based approach that is able to interpret regulations on underground utility infrastructure. Zhang and El-Gohary (2016) developed a tool for automated compliance checking that is based on the automated extraction of information from the international building code. In general, in the domain of building, much work has been done to automate the interpretation of building codes, as summarized by Fuchs (2021).

However, norms and regulations are not the only types of legal documents that are relevant for engineering practice. Particularly in commercial projects, contracts and their analysis are important tasks that can be supported by NLP. Again, in the domain of building much work has been done on the analysis of contracts, e.g. by Kim et al. (2020b) and Wong et al. (2024). Large manufacturers that have contractual relationship with many suppliers, e.g. in the automotive domain, often use highly standardized contracts for all of their supplies, so-called standard form contracts. The analysis of such contracts with NLP has been addressed e.g. by Braun (2021) and Lippi et al. (2019).

3.4.4 Engineering Education

LLMs and other NLP techniques already have a significant impact on education. While some of the impacts it has have been seen as negative, e.g. because the usage of LLMs by students presents a threat to the validity of examination (Morjaria et al. 2023), LLMs at the same time have the potential to improve (engineering) education. Vo et al. (2022), for example, showed how NLP can be used to help students to find the most suitable courses and jobs. Students also see LLMs as helpful in a wide variety of study-related tasks, including writing literature reviews and generating practice questions (Joshi et al. 2023).

NLP in engineering education cannot only support learners, but also teachers, for example in the assessment of exam questions, which could not only lead to a decreased workload for teachers but also result in more and faster feedback for students, ultimately supporting their learning process. While students in general seem to be open about the idea of being assessed by an AI, one important prerequisite for their acceptance is the possibility to challenge the AI assessment and have at least some human involvement at some point in the grading process (Bernabei et al. 2023; Braun et al. 2023).

While students have already accepted and embraced the usage of NLP technologies, many educational institutions often still struggle to find a way to address the use of AI in education. While some still try to outright prevent the use of AI, research shows that LLMs are already part of stu-

dents' daily routine: In a survey among undergraduate engineering students by Joshi et al. (2023), only 1% of respondents had never used ChatGPT, while more than 30% describe themselves as daily users and another 30% use it at least once a week. A complete ban on the usage of AI tools in education seems unlikely to be successful, not only because the tools are already so prevalent among students, but also because the detection of AI-generated submissions cannot be done reliably in an automated fashion, despite the availability of numerous commercial tools that claim to do just that (Bernabei et al. 2023; Fishchuk and Braun, 2023).

3.5 Outlook

Since the initial release of ChatGPT in November 2022, the field of NLP and particularly its applications, both in engineering and in general, have rapidly changed and continue to do so. The pace of innovation that we have seen in the last years makes it difficult to make sound predictions about the future developments in the field. Nevertheless, there are existing and emerging research streams that can give an indication of where the field is headed. In this section, we briefly want to look at five of such streams that already play an important role in NLP research and will have significant influence on the applicability of NLP systems in the years to come.

3.5.1 Local Inference

Training an LLM is a resource intensive endeavor. The open-source model BLOOM, for example, was trained on a total of 384 GPUs for about 3.5 months (Le Scao et al. 2023). The trend toward ever larger models will result in ever increasing hardware needs. While the increase in needed computing resources is particularly stark during the training phase, large models like Llama now also need 20 GB and more GPU memory at inference time, i.e. when being used after they have been trained initially. Very few users have such resources available in their local machine, let alone in a mobile device.

Recently, researchers started to focus on developing LLMs that are particularly resource-efficient during inference. Such models are designed to

be run on personal and even mobile devices. Such models could in the future not only help to bring down the costs of using the technology but also help to reduce latency and offer a privacy-preserving way of using LLMs because prompts that might contain sensible information will not have to leave the personal device anymore. While in many applications of ISE personal data might not be a large concern, trade secrets that are potentially sent to "the cloud," without having control over what happens with them once they leave the premises, certainly are.

3.5.2 Multimodality

As described in Section 3.3.2, the latest generation of models has already made a significant step toward multimodality. However, they are still mostly restricted to at most two modalities at a time, with one of them being text. Future generations of LLMs might be able to handle a wider variety of modalities including, but not limited to, texts, images, videos, and audio files, as both, input and output (Wu et al. 2023).

3.5.3 Transparency

Commercial LLMs, like the GPT-models of Open AI, are currently black boxes to their users. Not only because the way in which the models work is opaque, but also because it is often unclear on which data they have been trained, what happens to the information users put into the system, and how they come to the conclusions they put out. New legislation, like the AI Act of the European Union, will force providers of LLMs and other AI systems to create more transparency around their models. While some of the requirements formulated by the AI Act will simply force providers to disclose information they have not disclosed before, other requirements will need new technical approaches to explainability and transparency before they can be fulfilled. Therefore, this area will likely receive much attention in the coming months and years.

3.5.4 Retrieval-augmented Generation

As discussed in Section 3.2.1, the very size that allows LLMs to be so effective can also be a burden, for example, when it comes to re-training mod-

els on newer data or also fine-tuning them with not-publicly available documents. One method to compensate for that is to augment prompts to LLMs with external data. That can include structured data, like knowledge graphs, labeled examples, as in few-shot learning, but also unstructured textual data. These different approaches to using external in addition to the weights learned by the model to generate better outputs are summarized under the term retrieval-augmented generation. These technologies will play an important role in the future applications of LLMs, particularly in domains like engineering, where specialist knowledge is often needed.

3.5.5 Reinforcement Learning from Human Feedback

A widespread misconception about ML models is that they get better over time just by being used. In reality, an explicit feedback loop is needed that feeds user-feedback back into the training process of the model in order for the model to improve (or at least change) over time. Direct user-feedback, especially if collected at scale, can help to improve the quality of LLMs significantly (Shen et al. 2024). This process is referred to as reinforcement learning from human feedback (RLFH). However, the use of human feedback can also have a negative impact on LLMs. Malicious actors can use RLFH to change the responses of LLMs toward a certain direction (McIntosh et al. 2024). Even without malicious intentions, RLFH can still result in increasing the bias that models display (Liu 2023). Nevertheless, the potential of RLFH to improve LLMs will most likely lead to more research and also practical applications of the concept. Particularly in the context of private LLMs, i.e. models that are only accessible, for example, within a company or organization, RLFH can help to improve the output of LLMs. For public LLMs, research will first have to find ways to ensure only qualitative feedback is fed back into the system to avoid negative effects on the models.

References

Arora, C., Grundy, J., and Abdelrazek, M. (2023). Advancing requirements engineering through generative AI: assessing the role of LLMs. *arXiv preprint*: https://arxiv.org/abs/2310.13976.

Arora, C., Herda, T., and Homm, V. (2024). Generating test scenarios from NL requirements using retrieval-augmented LLMs: an industrial study. *arXiv preprint*: https://arxiv.org/abs/2404.12772.

Balloccu, S., Schmidtová, P., Lango, M., et al. (2024). Leak, cheat, repeat: data contamination and evaluation malpractices in closed-source LLMs. In *Proceedings of the 18th Conference OF the European chapter of the association for computational linguistics* (Vol. 1: Long Papers, pp. 67–93). Association for Computational Linguistics.

Bayraktar, D. and Gözlü, S. (1994). A knowledge-based expert system for technology acquisition in small and medium scale manufacturing organizations. *Computers & Industrial Engineering* 27 (1): 285–288. ISSN: 0360-8352. https://doi.org/10.1016/0360-8352(94)90291-7. 16th Annual Conference on Computers and Industrial Engineering.

Bender, E.M., Gebru, T., McMillan-Major, A., et al. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency, FAccT '*21 (pp. 610–623). Association for Computing Machinery. ISBN: 9781450383097. https://doi.org/10.1145/3442188.3445922.

Bernabei, M., Colabianchi, S., Falegnami, A., et al. (2023). Students' use of large language models in engineering education: a case study on technology acceptance, perceptions, efficacy, and detection chances. *Computers and Education: Artificial Intelligence* 5: 100172. ISSN: 2666-920X. https://doi.org/10.1016/j.caeai.2023.100172.

Braun, D. (2018). Automated semantic analysis, legal assessment, and summarization of standard form contracts. PhD thesis, Technische Universität München.

Braun, D., Faber, A., Hernandez-Mendez, A., et al. (2018). Automatic relation extraction for building smart city ecosystems using dependency parsing. In *NL4AI@ AI* IA* (pp. 29–39).

Braun, D., Klymenko, O., Schopf, T., et al. (2021). The language of engineering: training a domain-specific word embedding model for engineering. In *Proceedings of the 2021 3rd International Conference on Management Science and Industrial Engineering* (pp. 8–12).

Braun, D., Rogetzer, P., Stoica, E., et al. (2023). Students' perspective on AI-supported assessment of open-ended questions in higher education. In *CSEDU* (no. 2, pp. 73–79).

Brown, T., Mann, B., Ryder, N., et al. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems* 33: 1877–1901.

Calero, C., Ruiz, F., and Piattini, M. (2006). *Ontologies for Software Engineering and Software Technology*. Springer Science & Business Media.

Chiticariu, L., Li, Y., and Reiss, F. (2013). Rule-based information extraction is dead! long live rule-based information extraction systems! In *Proceedings of the 2013 conference on empirical methods in natural language processing* (pp. 827–832).

Devlin, J., Chang, M.-W., Lee, K., et al. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (Vol. 1, Long and Short Papers, pp. 4171–4186). Association for Computational Linguistics. https://doi.org/10.18653/v1/N19-1423.

Dwivedi, S., Ghosh, S., and Dwivedi, S. (2024). Navigating linguistic diversity: In-context learning and prompt engineering for subjectivity analysis in low-resource languages. *SN Computer Science* 5 (4): 418.

Fang, X., Che, S., Mao, M., et al. (2024). Bias of AI-generated content: an examination of news produced by large language models. *Scientific Reports* 14 (1): 1–20.

Fanni, S.C., Febi, M., Aghakhanyan, G., et al. (2023). *Natural Language Processing* (pp. 87–99). Springer International Publishing. ISBN: 978-3-031-25928-9. https://doi.org/10.1007/978-3-031-25928-9.5.

Fishchuk, V. and Braun, D. (2023). Efficient black-box adversarial attacks on neural text detectors. In *Proceedings of the 6th International Conference on Natural Language and Speech Processing* (*ICNLSP 2023*) (pp. 78–83). Association for Computational Linguistics.

Frieder, S., Pinchetti, L., Griffiths, R.-R., et al. (2024). Mathematical capabilities of chatgpt. *Advances in Neural Information Processing Systems* 36.

Fuchs, S. (2021). Natural language processing for building code interpretation: systematic literature review report. In *Proceedings of the Conference CIB W78* (*pp. 11–15*).

Greshake, K., Abdelnabi, S., Mishra, S., et al. (2023). Not what you've signed up for: compromising realworld llm-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security, AISec '23* (pp. 79–90). Association for Computing Machinery. ISBN: 9798400702600. https://doi.org/10.1145/3605764.3623985.

Gulla, J.A. (2009). Experiences with industrial ontology engineering. In *Enterprise information systems* (pp. 61–72). Springer Berlin Heidelberg. ISBN: 978-3-642-00670-8.

Joshi, I., Budhiraja, R., Tanna, P. et al. (2023). "With great power comes great responsibility!": Student and instructor perspectives on the influence of LLMs on undergraduate engineering education. *arXiv e-prints, pages arXiv–2309*.

Katz, D.M., Bommarito, M.J., Gao, S., et al. (2024). Gpt-4 passes the bar exam. *Philosophical Transactions of the Royal Society A* 382 (2270): 20230254.

Khurana, D., Koli, A., Khatter, K., et al. Natural language processing: state of the art, current trends and challenges. *Multimedia Tools and Applications* 82 (3): 3713–3744.

Kim, Y., Lee, J., Lee, E.-B., et al. (2020a). Application of natural language processing (NLP) and text-mining of big-data to engineering-procurement-construction (epc) bid and contract documents. In *2020 6th Conference on Data Science and Machine Learning Applications* (*CDMA*) (pp. 123–128). https://doi.org/10.1109/CDMA47397.2020.00027.

Kim, Y., Lee, J., Lee, E.-B., et al. (2020b). Application of natural language processing (NLP) and text-mining of big-data to engineering-procurement-construction (EPC) bid and contract documents. In *2020 6th Conference on Data Science and Machine Learning Applications* (*CDMA*) (pp. 123–128). IEEE.

Klymenko, O., Braun, D., and Matthes, F. (2020). Automatic text summarization: a state-of-the-art review. ICEIS (1) (pp. 648–655).

Kumara, S.R.T., Kashyap, R.L., Moodie, C.L. (1987). Expert system for industrial facilities layout planning and analysis. *Computers & Industrial Engineering* 12 (2): 143–152. ISSN: 0360-8352. https://doi.org/10.1016/0360-8352(87)90007-6.

Kung, T.H., Cheatham, M., Medenilla, A., et al. (2023). Performance of ChatGPT on USMLE: potential for AI-assisted medical education using

large language models. *PLoS Digital Health* 2 (2): e0000198.

Laurençon, H., Saulnier, L., Wang, T., et al. (2022). The bigscience roots corpus: a 1.6 tb composite multilingual dataset. *Advances in Neural Information Processing Systems* 35: 31809–31826.

Le Scao, T., Fan, A., Akiki, C., et al. (2023). Bloom: a 176b-parameter open-access multilingual language model.

Leo Kumar, S.P. (2019). Knowledge-based expert system in manufacturing planning: state-of-the-art review. *International Journal of Production Research* 57 (15–16): 4766–4790.

Li, Y., and Starly, B. (2024). Building a knowledge graph to enrich ChatGPT responses in manufacturing service discovery. *Journal of Industrial Information Integration* 40:100612. ISSN: 2452-414X. https://doi.org/10.1016/j.jii.2024.100612.

Lippi, M., Palka, P., Contissa, G., et al. (2019). Claudette: an automated detector of potentially unfair clauses in online terms of service. *Artificial Intelligence and Law* 27: 117–139.

Liu, G.K.-M. (2023). Perspectives on the social impacts of reinforcement learning with human feedback. *arXiv preprint*: https://arxiv.org/abs/2303.02891.

Liu, P., Qian, L., Zhao, X., et al. (2024). Joint knowledge graph and large language model for fault diagnosis and its application in aviation assembly. In *IEEE Transactions on Industrial Informatics* (pp. 1–10). https://doi.org/10.1109/TII.2024.3366977.

Marcus, G., Davis, E., and Aaronson, S. (2022). A very preliminary analysis of dall-e 2. *arXiv preprint*: https://arxiv.org/abs/2204.13807.

McIntosh, T.R., Susnjak, T., Liu, T., et al. (2024). The inadequacy of reinforcement learning from human feedback - radicalizing large language models via semantic vulnerabilities. In *IEEE transactions on cognitive and developmental systems* (pp. 1–14). https://doi.org/10.1109/TCDS.2024.3377445.

Meyer, L.-P., Stadler, C., Frey, J., et al. (2024). LLM-assisted knowledge graph engineering: Experiments with chatgpt. In *First working conference on artificial intelligence development for a resilient and sustainable tomorrow* (pp. 103–115). Springer Fachmedien Wiesbaden. ISBN: 978-3-658-43705-3.

Mikolov, T., Sutskever, I., Chen, K., et al. (2013). Distributed representations of words and phrases and their compositionality. *Advances in*

Neural Information Processing Systems 26.

Min, B., Ross, H., Sulem, E., et al. (2023). Recent advances in natural language processing via large pre-trained language models: a survey. *ACM Computing Survey* 56 (2): 1–40. ISSN: 0360-0300. https://doi.org/10.1145/3605943.

Morjaria, L., Burns, L., Bracken, K., et al. (2023). Examining the threat of ChatGPT to the validity of short answer assessments in an undergraduate medical program. *Journal of Medical Education and Curricular Development* 10: 23821205231204178. https://doi.org/10.1177/23821205231204178.

Motoki, F., Neto, V.P., and Rodrigues, V. (2024). More human than human: Measuring ChatGPT political bias. *Public Choice* 198 (1): 3–23.

Pal, S., Bhattacharya, M., Lee, S.-S., et al. (2024). A domain-specific next-generation large language model (LLM) or ChatGPT is required for biomedical engineering and research. *Annals of Biomedical Engineering* 52 (3): 451–454.

Pan, J.Z., Razniewski, S., Kalo, J.-C., et al. (2023). Large language models and knowledge graphs: opportunities and challenges.

Pennington, J., Socher, R., and Manning, C.D. (2014). Glove: Global vectors for word representation. In *Proceedings of the 2014 conference on empirical methods in natural language processing* (*EMNLP*) (pp. 1532–1543).

Poveda-Villalón, M., Fernández-Izquierdo, A., Fernández-López, M., et al. (2022). Lot: an industrial oriented ontology engineering framework. *Engineering Applications of Artificial Intelligence* 111: 104755. ISSN: 0952-1976. https://doi.org/10.1016/j.engappai.2022.104755.

Pursnani, V., Sermet, Y., Kurt, M., et al. (2023). Performance of ChatGPT on the US fundamentals of engineering exam: comprehensive assessment of proficiency and potential implications for professional environmental engineering practice. *Computers and Education: Artificial Intelligence* 5: 100183.

Raffel, C., Shazeer, N., Roberts, A., et al. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research* 21 (140): 1–67.

Rosadini, B., Ferrari, A., Gori, G., et al. (2017). Using NLP to detect requirements defects: An industrial experience in the railway domain. In *Requirements engineering: foundation for software quality: 23rd in-*

ternational working conference, REFSQ 2017, Essen, Germany, February 27–March 2, 2017, *Proceedings 23* (pp. 344–360). Springer.

Schneider, P., Schopf, T., Vladika, J., et al. (2022). A decade of knowledge graphs in natural language processing: A survey. In *Proceedings of the 2nd conference of the Asia-Pacific chapter of the association for computational linguistics and the 12th international joint conference on natural language processing* (Vol. 1: Long Papers, pp. 601–614). Association for Computational Linguistics.

Schopf, T., Braun, D., and Matthes, F. (2021). Lbl2vec: An embedding based approach for unsupervised document retrieval on predefined topics. In *Proceedings of the 17th international conference on web information systems and technologies - WEBIST* (pp. 124–132). INSTICC, SciTePress. ISBN: 978-989-758-536-4. https://doi.org/10.5220/0010710300003058.

Schopf, T., Weinberger, P., Kinkeldei, T., et al. (2022). Towards bilingual word embedding models for engineering: Evaluating semantic linking capabilities of engineering-specific word embeddings across languages. In *Proceedings of the 4th international conference on management science and industrial engineering* (pp. 407–413). Association for Computing Machinery. ISBN: 9781450395816. https://doi.org/10.1145/3535782.3535835.

Schopf, T., Braun, D., and Matthes, F. (2023a). Semantic label representations with lbl2vec: A similarity-based approach for unsupervised text classification. In *Web information systems and technologies* (pp. 59–73). Springer International Publishing.

Schopf, T., Braun, D., and Matthes, F. (2023b). Evaluating unsupervised text classification: Zero-shot and similarity-based approaches. In *Proceedings of the 2022 6th international conference on natural language processing and information retrieval, NLPIR '22* (pp. 6–15). Association for Computing Machinery. ISBN: 9781450397629. https://doi.org/10.1145/3582768.3582795.

Shen, W., Zhang, X., Yao, Y., et al. (2024). Improving reinforcement learning from human feedback using contrastive rewards. arXiv preprint: https://arxiv.org/abs/2403.07708.

Terkaj, W., Borgo, S., and Sanfilippo, E.M. (2022). Ontology for industrial engineering: a dolce compliant approach. In *FOMI*.

Touvron, H., Lavril, T., Izacard, G., et al. (2023). Llama: open and efficient foundation language models. arXiv preprint: https://arxiv.org/abs/2302.13971.

Trewartha, A., Walker, N., Huo, H., et al. (2022). Quantifying the advantage of domain-specific pre-training on named entity recognition tasks in materials science. *Patterns* 3 (4): 100488. ISSN: 2666-3899. https://doi.org/10.1016/j.patter.2022.100488.

Urchs, S., Thurner, V., Aßenmacher, M., et al. (2023). How prevalent is gender bias in ChatGPT? – Exploring German and English ChatGPT responses. *arXiv preprint*: https://arxiv.org/abs/2310.03031.

Valmeekam, K., Olmo, A., Sreedharan, S., et al. (2022). Large language models still can't plan (a benchmark for LLMs on planning and reasoning about change). *arXiv preprint*: https://arxiv.org/abs/2206.10498.

van Deemter, K. (2024). The pitfalls of defining hallucination. *Computational Linguistics* 50: 1–10.

Van Ruijven, L.C. (2013). Ontology for systems engineering. *Procedia Computer Science* 16: 383–392.

Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems* 30.

Veizaga, A., Shin, S.Y., and Briand, L.C. (2024). Automated smell detection and recommendation in natural language requirements. *IEEE Transactions on Software Engineering* 50: 695–720.

Vo, N.N.Y., Vu, Q.T., Vu, N.H., et al. (2022). Domain-specific NLP system to support learning path and curriculum design at tech universities. *Computers and Education: Artificial Intelligence* 3: 100042. ISSN: 2666-920X. https://doi.org/10.1016/j.caeai.2021.100042.

Wei, T., Feng, D., Song, S., et al. (2024). An extraction and novelty evaluation framework for technology knowledge elements of patents. *Scientometrics* 129: 1–26.

Wong, S., Zheng, C., Su, X., et al. (2024). Construction contract risk identification based on knowledge-augmented language models. *Computers in Industry* 157: 104082.

Wu, S., Fei, H., Qu, L., et al. (2023). Next-GPT: Any-to-any multimodal LLM. *arXiv preprint*: https://arxiv.org/abs/2309.05519.

Xu, X. and Cai, H. (2021). Ontology and rule-based natural language processing approach for interpreting textual regulations on underground

utility infrastructure. *Advanced Engineering Informatics* 48: 101288. ISSN: 14740346. https://doi.org/10.1016/j.aei.2021.101288.

Yin, S., Fu, C., Zhao, S., et al. (2023). A survey on multimodal large language models. *arXiv preprint*: https://arxiv.org/abs/2306.13549.

Yue, G., Liu, J., Hou, Y., et al. (2022). A novel patent knowledge extraction method for innovative design. *IEEE Access* 11: 2182–2198.

Zhang, J. and El-Gohary, N.M. (2016). Semantic NLP-based information extraction from construction regulatory documents for automated compliance checking. *Journal of Computing in Civil Engineering* 30 (2): 04015014.

Zhao, L., Alhoshan, W., Ferrari, A., et al. (2021). Natural language processing for requirements engineering: A systematic mapping study. *ACM Computing Surveys* 54 (3): 1–4. ISSN: 0360-0300. https://doi.org/10.1145/3444689.

Zheng, Z., Lu, X.-Z., Chen, K.-Y., et al. (2022). Pretrained domain-specific language model for natural language processing tasks in the AEC domain. *Computers in Industry* 142: 103733. ISSN: 0166-3615. https://doi.org/10.1016/j.compind.2022.103733.

Zhong, K., Jackson, T., West, A., et al. (2024). Natural language processing approaches in industrial maintenance: a systematic literature review. *Procedia Computer Science* 232: 2082–2097. ISSN: 18770509. https://doi.org/10.1016/j.procs.2024.02.029. 5th International Conference on Industry 4.0 and Smart Manufacturing (ISM 2023).

Zhou, Y., Lin, H., Liu, Y., et al. (2019). A novel method to identify emerging technologies using a semi-supervised topic clustering model: a case of 3D printing industry. *Scientometrics* 120: 167–185.